Code No: V3121/R07 **Set No:1**

III B.Tech I Semester Regular & Supplementary Examinations, November 2011

DIGITAL IC APPLICATIONS

(Electronics and Communications Engineering)

Time: 3 Hours **Max Marks: 80**

Answer any FIVE Questions

All Questions carry equal marks

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1. Explain about the steady state electrical CMOS behaviors for
   i) Resistive loads ii) Non ideal inputs [8+8]

2. a) What is the necessity of separate interfacing circuit to connect CMOS gate to TTL. Draw the interface circuit and explain the operation.
   b) Explain the following terms with reference to TTL gate. i) Logic levels ii)D.C noise margin iii)Low state unit loads iv)High state fan out. [8+8]

3. a) Write a VHDL entity and architecture for a 3-bit synchronous counter using flip-flops.
   b) Discuss the steps in VHDL design flow. [8+8]

4. Design the logic circuit and write a data-flow style VHDL program for the following functions?
   a) $F(A)= \Pi_{p,q,r,s}(1,3,4,5,6,7,9,12,13,14)$
   b) $F(X) = \Sigma_{A.B.C.D}(3,5,6,7,13) + d(1,2,4,12,15)$ [8+8]

5. a) Design a priority encoder for 16 inputs using two 74 X 148 encoders.
   b) Design a 24-bit group ripple adder using 74 X 283 IC. [8+8]

6. a) Design a barrel shifter for 8-bit using three control inputs. Write a VHDL program for the same in data flow style.
   b) Write a behavioral VHDL program to compare 16-bit signed and unsigned integers. [8+8]

7. a) Discuss the logic circuit of 74 X 377 register. Write a VHDL program for the above logic.
   b) Design an Excess-3 decimal counter using 74 X 163 and explain the operation with the help of timing waveforms. [8+8]

8. a) Design a 8X4 diode ROM using 74X138 for the following data starting from the first location. B, 2, 4, F, A, D, E, F.
   b) With a neat diagram explain the general architecture of CPLD. Discuss the key features of Xilinx XC9500 CPLD family. [8+8]

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Code No: V3121/R07 **Set No:2**

III B.Tech I Semester Regular & Supplementary Examinations, November 2011

DIGITAL IC APPLICATIONS

(Electronics and Communications Engineering)

Time: 3 Hours **Max Marks: 80**

Answer any FIVE Questions

All Questions carry equal marks

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1. a) What are the advantages and disadvantages of CMOS technology? Discuss about the recent developments in the process technology.
   b) Draw the CMOS circuit diagram of tri-state buffer. Explain the circuit with the help of logic diagram and function table. [8+8]

2. a) Explain sinking current and sourcing current of TTL output. Which of the above parameters decide the fan-out and how.
   b) Design a transistor circuit of 2-input ECL NOR gate. Explain the operation with the help of function table. [8+8]

3. a) Write a VHDL Entity and Architecture for the following function. $F(x) = a \oplus b \oplus c$ also draw the relevant logic diagram.
   b) Explain the use of packages. Give the syntax and structure of a package in VHDL. [8+8]

4. Design the logic circuit and write a data flow style VHDL program for the following function. $F(Q)=\Sigma_{A,B,C,D}\,(0,2,5,7,8,10,13,15)+d(11)$
   $F(Q)=\Sigma_{A,B,C,D}\,(1,4,5,7,9,10,13,15)+d(11)$ [16]

5. Design a 16-bit ALU using 74 X 381 and 74 X 182Ic's. Implement VHDL source code using data flow style for the same. [16]

6. a) Explain the operation of simple floating point encoder.
   b) Design a 4 X 4 combinational multiplier and then write the necessary VHDL program data flow model. [8+8]

7. a) Design a conversion circuit to convert a T-flip-flop to JK-flip-flop.
   b) Design a 8-bit synchronous binary counter with serial enable control. [8+8]

8. a) What is the difference between SRAM and ROM. Give in detail the description of 4X3 static ROM.
   b) With the help of timing waveforms, explain the read and write operations of static RAM. [8+8]

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Code No: V3121/R07 **Set No: 3**

III B.Tech I Semester Regular & Supplementary Examinations, November 2011

DIGITAL IC APPLICATIONS

(Electronics and Communications Engineering)

Time: 3 Hours **Max Marks: 80**

Answer any FIVE Questions

All Questions carry equal marks

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1. a) Draw the resistive model of a CMOS inverter and explain its behavior for LOW and HIGH outputs.
   b) Explain the following terms with reference to CMOS logic i) Logic levels ii) power supply rails iii) DC noise margin iv) Propagation delay [8+8]

2. a) Mention the DC margin noise levels of ECL 10 K family.
   b) List out TTL families and compare them with reference to propagation delay, power consumption, speed, power product and low level input current. [8+8]

3. a) What are Sub-Programs? Give a detail view of functions and Procedures.
   b) Explain the Implicit and Explicit visibility of a library in VHDL. [8+8]

4. a) Explain the behavioral design model of VHDL.
   b) Design the logic circuit and write a data flow style VHDL program for the following function. $F(R) = \pi_{A,B,C,D}(1,4,5,7,9,13,15)$ [8+8]

5. a) Design a full adder using two half adders. Write VHDL data flow program for the above implementation.
   b) Design a 16-bit ALU using 74 X 381 and 75 X 182 ICs. [8+8]

6. a) What is a comparator? Draw and explain the structure and working of a comparator.
   b) Write a VHDL program for the circuit that counts number of ones in a 16-bit register using structural style of modeling. [8+8]

7. a) Distinguish between latch and flip-flop. Show the logic diagram for both. Explain the operation with the help of function table.
   b) Design a 3-bit LFSR counter using 74 X 194. List out the sequence assuming the initial state is 111. [8+8]

8. a) Discuss in detail ROM access mechanism with the help of timing waveforms.
   b) Explain how PROM, EPROM and EEPROM technologies differ from each other. [8+8]

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Code No: V3121/R07

Set No:4

III B.Tech I Semester Regular & Supplementary Examinations, November 2011

DIGITAL IC APPLICATIONS

(Electronics and Communications Engineering)

Time: 3 Hours **Max Marks: 80**

Answer any FIVE Questions

All Questions carry equal marks

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1. a) Design a 4 input CMOS OR-AND INVERT state. Explain the circuit with the help of logic diagram and function table.
   b) Compare different logic families [8+8]

2. a) Design a TTL three state NAND gate explain the operation.
   b) List out different categories of characteristics in a TTL data sheet. Discuss electrical and switching characteristics. [8+8]

3. a) Write down the VHDL program for comparing 8-bit unsigned integer.
   b) Give an account of package declaration and package body. [8+8]

4. a) Explain the data-flow design elements of VHDL.
   b) Design the logic circuit and write a data flow style VHDL program for the following function. $F(p) = \Sigma_{A,B,C,D}(2,6,7,9,11,15)$. [8+8]

5. a) Write a VHDL program for 74 X 245 IC.
   b)Using two 74X138 decoders design a 4 to 16 decoder. [8+8]

6. Explain the operation of barrel shifter and write a VHDL. Program for 16 bit barrel shifter for left and right circular shifts. [16]

7. a) Draw the logic diagram of 74 X 194 and explain the operation.
   b) Give a VHDL code for a 4-bit up-counter with enable and clear inputs. [8+8]

8. a) Draw the block diagram of synchronous SRAM and explain each one of it precisely.
   b) Explain the internal structure of 64K X 1 DRAM. With the help of timing waveforms describe the DRAM access. [8+8]

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